G12Adv	Mathematics	الرياضيات	T1 - 2022/23

Multiple Choice Part	جزء الاختيار من متعدد
Show all your work when answering these questions.	يجب كتابة خطوات الحل التفصيلية للمفردات الاختبارية كافة.

Multiple Choice — اختيار من متعدد

1. Find $y'(x)$ implicitly for $x^2y - 3y^3 = x^2 + 1$.

A) $y'(x) = \dfrac{2x^2 + 2xy}{x^2 + y^2}$

B) $y'(x) = \dfrac{2x - 2y}{x^2y + 9y^2}$

C) $y'(x) = \dfrac{2x - 2xy}{x^2 + 9y^2}$

D) $y'(x) = \dfrac{2x - 2xy}{x^2 - 9y^2}$

2. Find the derivative of the function $f(t) = 3t^{\pi} - 2t^{1.3}$.

A) $f'(t) = 3\pi - 2.6t$

B) $f'(t) = 3\pi t - 2.6t$

C) $f'(t) = 3\pi t^{\pi-1} - 2.6t^{0.3}$

D) $f'(t) = 3\pi t^{\pi^2} - 2.6t^{0.3}$

3. Use L'Hôpital's rule to compute the limit.

$$\lim_{x\to 0} \frac{\sin x - x}{x^3}$$

A) 0

B) $-\dfrac{1}{6}$

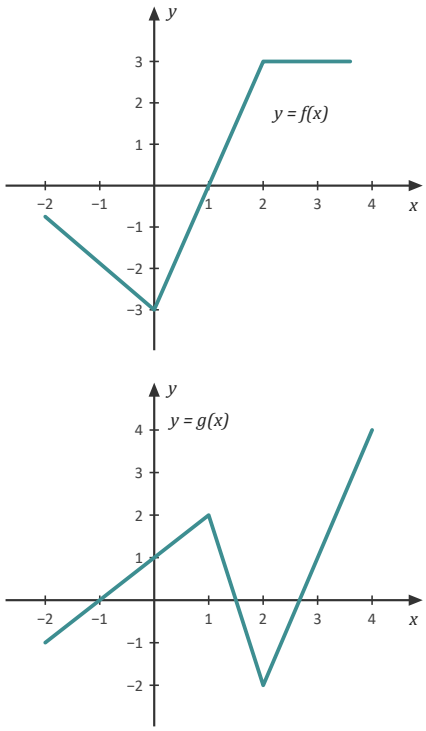
C) 8

D) 6

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Multiple Choice — اختيار من متعدد

4. **Use the graphs to find the derivative of the composite function $f(g(x))$ at $x = 3$.**



A) 0 **B)** 3

C) 9 **D)** Does not exist

5. **If $f(x) = \dfrac{9 - x^2}{mx + 2}$ is continuous on $(-\infty, \infty)$, find the value of m, where m is a constant number.**

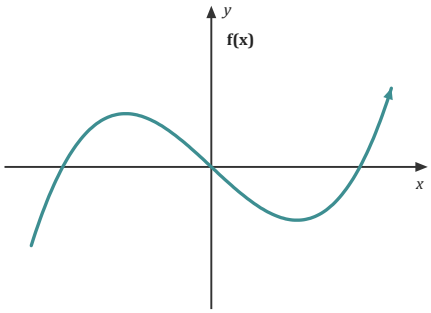
A) -1 **B)** 0

C) 1 **D)** 9

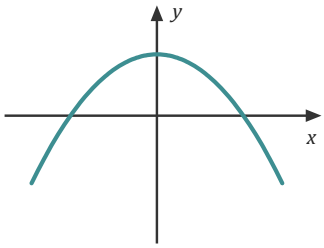
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Multiple Choice	اختيار من متعدد

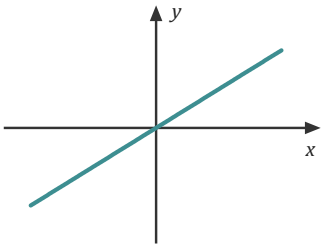
6. **Use the graph of $y = f(x)$ to sketch a graph of $y = f'(x)$.**



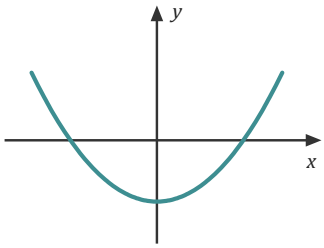
A)



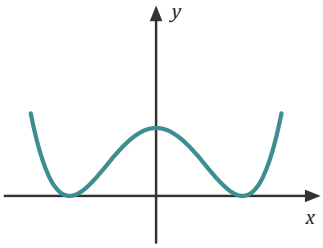
B)



C)



D)



7. **Find the derivative of the function $f(x) = \sin^2(\tan x)$.**

A) $f'(x) = 2\sin(\tan x) \cdot \cos(\tan x) \cdot \sec^2 x$

B) $f'(x) = 2\sin x \cdot (\tan x) \cdot \cos x + \sec^2 x \cdot \sin^2 x$

C) $f'(x) = 2\sin(\tan x) \cdot \sec^2 x$

D) $f'(x) = 2\sin(\tan x) \cdot \cos x \cdot \sec^2 x$

8. **Evaluate**

$$\lim_{x \to 0} x^2 \csc^2 x$$

A) 0 **B)** 1

C) 2 **D)** Does not exist

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Multiple Choice	اختيار من متعدد

9. **Evaluate**

$$\lim_{x\to 1} \frac{x+\sqrt{x}-2}{x-1}$$

A) 0

B) $\frac{3}{2}$

C) 2

D) $\frac{2}{3}$

10. **Find all values of x for which the tangent line to the curve $y = x^4 - 2x^2 + 3$ is horizontal.**

A) $x = -\sqrt{2},\ x = \sqrt{2}$

B) $x = 1,\ x = \sqrt{3}$

C) $x = -1,\ x = 0,\ x = 1$

D) $x = -1,\ x = 2$

11. **Evaluate**

$$\lim_{x\to\infty} \left(\sqrt{x^2+1} - x\right)$$

A) 0

B) 1

C) $\frac{1}{2}$

D) ∞

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12. **Find the derivative of the function** $f(x) = \sin^{-1}(\sqrt{x})$.

A) $f'(x) = \dfrac{1}{2\sqrt{x(1-x)}}$

B) $f'(x) = \dfrac{1}{2\sqrt{1-x^2}}$

C) $f'(x) = \dfrac{-1}{\sqrt{x(1-x)}}$

D) $f'(x) = \dfrac{1}{\sqrt{\sqrt{x}}\,(1-x)}$

13. **Evaluate**

$$\lim_{x\to 1} \frac{x^3-1}{x^2+2x-3}$$

A) 0

B) $\dfrac{1}{4}$

C) $\dfrac{1}{2}$

D) $\dfrac{3}{4}$

14. **Find the derivative of the function** $f(x) = \dfrac{\sqrt{\ln x}}{x}$.

A) $f'(x) = \dfrac{1-2\ln x}{\sqrt{\ln x}}$

B) $f'(x) = \dfrac{1}{2x^2\sqrt{\ln x}}$

C) $f'(x) = \dfrac{1-2\ln x}{2x^2\sqrt{\ln x}}$

D) $f'(x) = \dfrac{1-2\ln x}{2x^2}$

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15. Evaluate $\lim_{x\to 0} \sin\left(e^{-1/x^2}\right)$ if it exists.

A) -1 **B)** 0

C) 1 **D)** Does not exist

16. Find all horizontal asymptotes of $f(x) = \dfrac{x}{\sqrt{x^2+1}}$.

A) $y = -1$ **B)** $y = 0$

C) $y = 1$ **D)** $y = -1,\ y = 1$

17. Find an equation of the tangent line to the function $f(x) = \dfrac{2}{x+1}$ at $x = 1$.

A) $y = -\frac{1}{2}(x-1) + 1$

B) $y = -\frac{1}{2}(x+1) - 1$

C) $y = -2(x-1) + 1$

D) $y = \frac{1}{2}(x-1) - 1$

18. Use logarithmic differentiation to find the derivative of $f(x) = (\sin x)^x$.

A) $f'(x) = (\sin x)^x \cdot [x\cot x + \ln(\sin x)]$

B) $f'(x) = (\sin x)^x \cdot [x + \ln(\sin x)]$

C) $f'(x) = (\sin x)^x \cdot (x\cot x)$

D) $f'(x) = x\cot x + \ln(\sin x)$

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19. **Suppose that the height of a falling object t seconds after being dropped from a height of 64 ft is given by $s(t) = 64 - 16t^2$ ft. Find the average velocity between $t = 1$ and $t = 2$.**

A) -24 ft/s

B) -6 ft/s

C) -36 ft/s

D) -48 ft/s

20. **Given that $\lim_{x\to 0} \frac{\sin x}{x} = 1$, evaluate $\lim_{x\to 0} \frac{\cos^2 x - 1}{2x^2}$.**

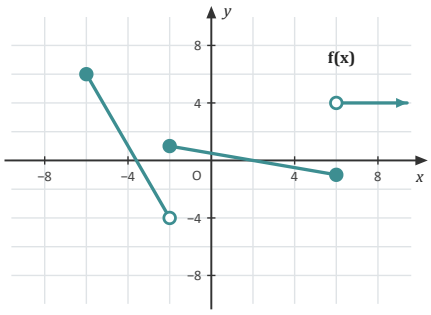
A) $-\frac{1}{2}$

B) $\frac{1}{2}$

C) 1

D) 2

21. **Use the graph to determine $\lim_{x\to -2} f(x)$, if it exists.**



A) -4

B) 1

C) 0

D) Does not exist

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22. Estimate the arc length of the curve $y = x^3 + 2$ on the interval $-1 \leq x \leq 1$ using $n = 2$ line segments.

A) $2\sqrt{2}$ B) $\sqrt{2}$

C) $\sqrt{6}$ D) 2

23. Find a value of c as guaranteed by the Mean Value Theorem for the function $f(x) = x^2 - 2x$ on the interval $[0, 2]$.

A) $C = 2$ B) $C = 0$

C) $C = 1$ D) $C = 1.5$

24. Find the linear approximation to $f(x) = \dfrac{2}{x}$ at $x_0 = 1$.

A) $L(x) = 4 - 4x$ B) $L(x) = 2x - 2$

C) $L(x) = 4 - 2x$ D) $L(x) = 6 - 2x$

25. Determine the interval(s) where $f(x) = \dfrac{\ln(1 - x^2)}{\sqrt{x}}$ is continuous.

A) $(1, \infty)$ B) $(0, \infty)$

C) $(-1, 1)$ D) $(0, 1)$